Bhumi Lipi

Master Plan v3: merged and final

LandTech Innovation Hackathon 2026 · Track: Digital Land Records, Verification & Digitization · Main problem: **P2 Deed & Ownership Verification** (P3, P4, P5 as supporting features) · Written Wed 14 Oct 2026 · Round 1 due **Fri 16 Oct 2026**

What this document is	The complete plan for the project: what exists, what was merged from the second plan you pasted, the important Bangladesh problems, international technology adapted at low cost, architecture, front end for every age, schedule, Antigravity prompts, submission kit, honest claims, risks. Bangla words are written in English letters because this PDF has no Bengali font; the app itself is fully Bangla and English.
How to use it	1) Read section 1 (what changed) and section 8 (what to do before Friday). 2) Run the prompts in section 9 one at a time. 3) Phase B (sections 5 and 8) is only for the Final Round, if selected.
What is NOT promised	A 10/10 score. The Judging Rubric is a separate document that has not been seen. Every number about cost savings is deliberately left out because nobody has measured it.

0. Status today (all tested)

Area	What works now	Proof
Core (P2)	Signed hash chain (SHA-256 + Ed25519, one key per office), append-only DB triggers, chain guard, QR seal, file fingerprint check, three verify results (Genuine, Outdated, Forged)	53 automatic tests
Fraud (P3)	11 explainable rules, risk score, fake-seller and double-sale blocks, liens and double-mortgage block, alerts	unit + HTTP tests
Mutation (P4)	Auto-start after registration, 8 automatic checks, tracking ID, officer queue, REVERSAL blocks, **new today: waiting-days aging chip**	unit + browser test
Citizen (P5)	Green / Amber / Red report, history, survey table, print, **new today: Simple mode** (three-word shape verdicts, big text, high contrast, read-aloud when a Bangla voice exists, ana-gonda calculator)	browser test
Trust	Public Merkle anchor, in-browser proof check, independent verifier script, attack lab (5 attacks on a copy)	verifier script + tests
Delivery	Concept note PDF (3 pages), README, SPEC, UI_SPEC, QA, threat model, demo script, Dockerfile, render.yaml, CI	in zip
NOT built yet	Seal stamp + verdict reveal animation, chain-crack view, diff viewer, mobile tab bar, shares, heirs, maps, OCR, offline QR, identity, SMS	plan only

1. The two plans merged: what was kept, changed or dropped

The second plan you pasted has good ideas, but parts of it do not match the real project or contain mistakes. Following it literally would have broken the working code or cost marks in Q&A.; Decisions:

Topic	Pasted plan said	Merged decision	Why
Main problem	P2, with P3/P4/P5 and P1	P2 main. P3/P4/P5 as one lifecycle. P1 OCR only in Phase B	Rule book: pick one and go deep
File names	app.py, crypto.py, static/, tests/test_unit.py	Keep the real names: api.py, chain.py, web/, tests/test_*.py	Renaming files two days before the deadline is pure risk
Block fields	seq, plot_id, kind, frm, to, price, doc_hash, prev_hash, ts, office_id	Keep frozen: plot_id, seq, kind, deed_no, frm, to, ts, doc_hash, price_bdt, office_id, prev	Changing them invalidates every signature and test
Block kinds	GENESIS, SALE, MUTATION, LIEN, LIEN_RELEASE, SUCCESSION	Real kinds: INIT, SALE, REVERSAL. Liens stay in their own table. SUCCESSION can be added later as a new kind	Additive only
Rule names and weights	KHAS_LAND, RAPID_RESALE 40, SURVEY_AREA_MISMATCH 45 ...	Real codes: GOV_LAND, RAPID_TRANSFER 30, SURVEY_MISMATCH 40 ... New rules use the same style	The code, tests and docs already use these
API	GET /api/verify/{hash}; POST /api/attack-lab/tamper and /restore	Real API: POST /api/verify, POST /api/lab/run (SPEC section 10)	The lab must never tamper the live database. It runs on a private copy

Topic	Pasted plan said	Merged decision	Why
Ana-gonda	"16 Ana = 64 Gonda = 100% share"	**Wrong.** 16 ana = the whole; 1 ana = 20 gonda; 1 gonda = 4 kora; 1 kora = 3 kranti. So the whole = 320 gonda. Built and tested with this rule	A wrong conversion would embarrass the team in front of Bangladeshi judges. Ask a land-office clerk to confirm
"30 to 45 day disconnect"	Stated as a fact	Removed	No source. Say "a gap between registration and mutation" instead
"2PC lock", "100/100", "tamper-proof"	Used as selling words	Say: status lock until mutation is approved; tamper-evident; detected by anchor and verifier	Wrong terms invite questions you cannot answer
Header badge	"Sec 8 Compliant"	"100% synthetic data" (already in the demo bar)	Only the organizers can judge compliance
Timeline	Start Thu 15 Oct	Start now (Wed 14 Oct). Simple mode and aging are already done	Only two days remain
Prompt A1	Palette + badge + simple mode + calculator in one prompt	Split into small prompts (section 9)	Small diffs the students can actually explain
Logo colors	#0F3B24, #1B5E3A, #C69A35, #F4EFE0	Keep the tokens sampled from the logo file (#0f3a1f, #b8963c on #f6f2e4)	Already tested for contrast in light and dark
Tests count	51+	53 (aging + share maths added)	Current number

Kept from the pasted plan: the share ledger and "oversell above 100 percent" idea, succession (warish) mutation, survey lineage graph, offline-signed QR, hybrid OCR with human review, 3-office witness signing, the synthetic Bangla data generator, a land tax check, vested / waqf flags, and the idea of three modes for different users.

2. Important Bangladesh problems solved inside the rule book

#	Problem (Bangladesh)	Rule book	Solution	Status
1	Forged or altered deeds	P2	Digital seal, signed chain, QR, file fingerprint	Built
2	Same land sold twice before mutation	P3	Plot lock until mutation approved; DB chain guard	Built
3	Registration and mutation in separate offices; delays	P4	Mutation starts automatically; 8 checks; tracking ID; timeline	Built
4	Mutation waiting with no accountability	P4	Waiting-days chip (amber 14, red 30 days)	**Built today**
5	Buyers depend on brokers; cannot check	P5	Plain-language check page, QR, print	Built
6	Low literacy, elders, low digital skill	P5	Simple mode: shape verdicts, big text, contrast, read-aloud, ana-gonda helper	**Built today**
7	Double mortgage	P3	One active lien per plot; lien blocks sale	Built
8	Government land sold: khas, acquisition, vested, waqf	P3, P5	Block. Khas and acquisition built; vested/waqf flags small add-on	Part built
9	Court case or stay order hidden	P5	Case flag built; add case number and stay order	Part built
10	CS/SA/RS/BS area mismatch	P3	Area warning built; lineage graph with shares in Phase B	Part built
11	Several co-owners; sale above own share	P3	Share ledger; SHARE_EXCEEDED blocks (F1)	Phase B
12	Inheritance (warish) mutation	P4	Succession workflow, shares sum to 1 (F2)	Phase B
13	Fake identity or fake power of attorney	P3	Demo identity token hash; POA forces officer review (F7)	Phase B
14	Land development tax arrears	P4	Ninth automatic check (F8)	Phase B
15	Boundary overlap, encroachment	P3	Polygon overlap rule + offline SVG mouza map (F4)	Phase B
16	Old paper khatian and deeds	P1	Hybrid Bangla OCR with human review (F6)	Phase B
17	No internet in villages	P2, P5	Offline-verifiable signed QR, PWA, SMS/USSD simulator (F5, F9)	Phase B
18	Insider tampering	P2	Public anchor, independent verifier, attack lab; witness co-signing (F11)	Built + F11
19	Courts and lawyers need evidence	P2	Printable title report with QR and proof path	Built

Legal rules for shares, heirs and tax are shown as **demo rules**. Have a land-law teacher or a land-office clerk confirm them before you call them correct.

3. International land technology adapted at low cost

Descriptions below come from public information about these systems. Verify each fact before quoting it in the concept note.

System	Idea	Gap for Bangladesh	Bhumi Lipi (cost-efficient)
Georgia	Fingerprints of land titles timestamped on a public blockchain	Chain fees and complexity	Office signatures + hash chain; only a periodic Merkle root is published. No per-transaction fee
Estonia	One secure layer between offices; hash-tree integrity proofs	Sub-registry and AC Land are separate; needs national ID	One lifecycle and one API; role-based; demo ID tokens now
India	One ID per land parcel (ULPIN); drone maps (SVAMITVA)	Many co-owners and old khatians per plot	One plot ID that deeds, surveys and shares attach to; free SVG mouza map later
Rwanda	Systematic registration of every parcel	Overlaps found late	Area and overlap rules at registration time
Sweden / UK	Blockchain and digital-street pilots	Registration and mutation are separate	Auto-start mutation and lock the plot between them
Zambia and others	Mobile-first access in low-connectivity places	Smartphones and data are uneven	Simple mode, voice, print, assisted use at union digital centers, SMS/USSD simulator

System	Idea	Gap for Bangladesh	Bhumi Lipi (cost-efficient)
Global cloud OCR	Digitize old records	Per-page fees at national scale	OpenCV + Tesseract Bangla first; human review for unclear fields

Honest cost drivers: hosting, database, hardware keys per office, officer training, and the people-time to digitize old records. **Near zero:** per-transaction fees (none), software licences (open source), publishing the anchor.

4. Architecture: build once, extend without rewriting

Real folder layout (keep it): `bhumi/` (chain.py, db.py, rules.py, mutation.py, service.py, seed.py, api.py) · `web/` (index.html, css/, js/, js/views/, assets/) · `tools/` (verify_anchor.py, make_demo_files.py) · `tests/` · `docs/` · server.py, Dockerfile, render.yaml.

Layer	What it does	Notes
Browser (vanilla JS, no build)	Check, Verify, Registrar, Mutation, Bank, Integrity, Lab, About; Simple mode; QR encoder written in-house and decoded with OpenCV in tests	Text only via textContent; strict CSP
HTTP layer (Python standard library)	Routes, demo roles, rate limit, 64 KB body cap, security headers, static files, path-traversal safe	One dependency in total: cryptography
Service layer (framework-free)	register_sale, check_plot, verify, decide, liens, integrity, anchor, lab	Moving to FastAPI later changes only api.py
Crypto and rules	SHA-256, Ed25519 per office, Merkle tree; 11 rules with Bangla and English text	Pure functions, easy to test and explain
Database (SQLite, PostgreSQL later)	Append-only triggers on blocks, anchors, mutation_events, audit_log; chain guard trigger	Rules enforced by the database itself

Frozen block format and extension points

- **Frozen canonical fields:** plot_id, seq, kind, deed_no, frm, to, ts, doc_hash, price_bdt, office_id, prev (JSON, sorted keys, compact, UTF-8). hash = SHA-256; sig = Ed25519 over the hash bytes. New facts (for example area sold) go into the deed document text, whose fingerprint doc_hash is already inside the signed block.
- **Additive database changes only:** new columns with defaults, new tables, a schema_version row and numbered migrations. No renames, no drops.
- **Rules as a registry (task B0):** each rule is (code, level, weight, bn, en, check). Risk score, alerts, mutation checklist and UI reasons read the same list, so a new rule is one entry plus one test.
- **Feature flags:** env BHUMI_FEATURES (for example shares,heirs,map). Unfinished features ship switched off and cannot break the demo.
- **Independent verifier** never imports the app and ignores unknown fields, so exports stay checkable as the system grows.
- **Tests are the contract:** every feature adds unit tests, API tests and a browser-story line. A feature is done only when all earlier tests still pass.

Real API (all JSON under /api)

GET plots, plots/{id}, stats, integrity, anchor, anchor/proof/{hash}, export, keys, liens, mutations/{tracking}, mutations/queue, deeds/{hash}, deeds/{hash}/document, alerts, health · POST verify, risk, deeds, mutations/{tracking}/decision, liens, liens/{id}/release, lab/run. Roles by header X-Demo-Role (demo only, not real login). Errors always {error:{code, bn, en}}. Details: docs/SPEC.md section 10.

Phase B data model (additive)

plots: vested, waqf, tax_paid_until, polygon, case_no, stay_order · holders(plot_id, holder, share_num, share_den, since_seq) · heirs(mutation_id, holder, share_num, share_den) · survey_links(...) · identity_attestations(deed_hash, token_hash, method, ts) · docs_ocr(...) · witness_signatures(anchor_id, office_id, sig).

Planned new rules (flagged): SHARE_EXCEEDED (block), CO_OWNERS_CONSENT (warn 50), HEIRS_SUM_INVALID (block), SURVEY_SHARE_BREAK (warn 45), BOUNDARY_OVERLAP (warn 45), IDENTITY_REQUIRED (block), POA_SALE (warn 40), VESTED_OR_WAQF (block), TAX_DUE (warn 35).

5. Front end, brand and "easy for every age"

Element	Specification
Brand tokens (from the logo)	Deep green #0f3a1f, gold #b8963c, cream #f6f2e4; dark theme gold on deep green. Verdict colors always paired with a shape and a word
Typography	Hind Siliguri (Bangla + Latin), Fraunces for Latin headings, monospace for hashes. Bangla line-height at least 1.7, no letter-spacing, no italics. Self-host fonts (planned) so Bangla never depends on Google Fonts
Three modes	Simple (citizen default), Standard (full report with proof), Officer (registrar, AC Land, bank)
Simple mode (built)	Three-word verdict with shapes: shield "Egon / Go ahead", triangle "Sabdhan / Careful", octagon "Thamun / Stop". Text size 100 / 125 / 150 percent. High-contrast theme. Listen button reads a Bangla summary when the device has a Bangla voice (hidden otherwise, never autoplays). Print and Share buttons. Ana-gonda to decimal calculator. Full details one tap away
Animation (purposeful)	Seal stamp, chain link, lock, QR scan, verdict reveal under 1.4 s, skippable, only transform and opacity, nothing moves with reduced-motion. Remaining to build: seal stamp and reveal, chain crack, diff viewer, hash avalanche
Mobile	Bottom tab bar (planned), 44 px targets, no sideways scroll at 360 px (tested)
Accessibility	Skip link, landmarks, labels, visible focus, live regions for results, contrast tested in light and dark, keyboard access
Performance targets (not yet measured)	First load under 150 KB without fonts, LCP under 2 s on slow 4G, no layout shift

Limits to say out loud: the Listen button depends on the phone having a Bangla voice, so test it on the demo phone before you promise it. The ana-gonda helper is an estimate, not a legal calculation.

6. Plan

Phase A: now to Friday (submit what works)

When	Do	Hours	Stop rule
Wed night 14 Oct	Run section 9 prompt A0 (tag + test baseline). Then A3: Seal component and verdict reveal. Review with the strict-teacher prompt. Commit	4	If tests fail and are not fixed in 30 min, go back to the tag
Wed night (optional)	A4: chain crack + diff viewer + hash avalanche. Mobile tab bar	4	Skip if tired. Simple mode already gives the main "all ages" value
Wed midnight	**Feature freeze.** No new features after this	0	Hard rule
Thu 15 Oct	Deploy to Render; test on a phone with mobile data; fill team and mentor names in the concept note; README (live link, every AI tool and what it did); record the 3-minute video; run /rubric-check with the official rubric; /pre-submit; rehearse docs/QA.md aloud	8	Submit-quality day, no coding
Fri 16 Oct	Final test on the live link. **Submit before noon.** Screenshot the confirmation	1	Do not wait for the last hour

Phase B: after selection (Final Round date to be announced)

Slot	Features (each one ticket, tests first, behind a flag)
B1	B0 extension points (rule registry, features.py, migrations), F1 share ledger, F8 tax check + vested/waqf flags
B2	F3 survey lineage graph with shares, F2 succession mutation
B3	F4 offline SVG mouza map + overlap rule, F5 offline-signed QR + PWA
B4	F6 hybrid Bangla OCR (P1), F7 identity token + power-of-attorney, F11 witness co-signing, F9 SMS/USSD simulator, F10 data generator, optional PostgreSQL

Final Round rules: update the repo and README "Changes since Round 1"; 10-slide pitch deck PDF; 5 minutes pitch and live demo, 3 minutes questions; **only the two students speak**. Choose two hero features for the pitch (suggestion: share ledger and offline QR, or the mouza map) and keep the rest on one roadmap slide.

7. Antigravity prompts (copy and paste, one per conversation)

Use Planning mode: approve the plan before the agent edits. Antigravity reads AGENTS.md, GEMINI.md and .agent/. After every prompt run /test-all, read the diff, run /explain-back on anything you cannot explain, then commit. Never let the agent edit tests to make them pass. The project already contains: the Simple-mode prompt (done) and the aging prompt (done), so start at A0.

A0: safety net (do first)

```
Read AGENTS.md and docs/SPEC.md. Do not change any code. Run the unit tests and the browser test and tell me the exact results. Then create a git tag named round1-freeze on the current commit. Report the tag and the test summary.
```

A3: Seal component and verdict reveal (front end only)

```
Read AGENTS.md, docs/UI_SPEC.md and docs/WEBSITE_PLAN.md sections 3, 4 (The Seal) and 5.3 (verdict reveal choreography). Front end only, no changes under bhumi/. Build a Seal component (SVG, 160 px and 48 px) with variants genuine (green shield), outdated (amber, faded), forged (red, cracked) and pending (gold, dashed). Use it in the verdict banner on the Check report and on Verify results. Add the reveal: a gold scan line sweeps the card (360 ms), then the Seal stamps in (520 ms, scale 1.6 to 1 with -8 degree rotation), then the text slides up. Total under 1.4 s; a click or Esc skips it; with prefers-reduced-motion show the final state instantly. Animate only transform and opacity. Keep every existing id, class and label used by tests/e2e_playwright.py (the banner keeps class verdict). Add browser-test steps. Show a plan first and wait for approval. Then run all tests and report.
```

A4 (optional): chain crack, diff viewer, hash avalanche

```
Read docs/WEBSITE_PLAN.md sections 4 (chain view, diff viewer) and 5.4 (Try to cheat). Front end only. (1) In the ownership history, draw a red broken-link icon and tint the later cards when the API says chain.ok is false. (2) In Verify > Try to cheat, show a character-level diff between the original deed text and the edited text, and below it the two SHA-256 fingerprints with the differing characters highlighted and a count such as "1 character changed, 62 of 64 fingerprint characters changed" (computed in the browser). Respect reduced motion, Bangla and English, textContent only. Do not break existing tests. Plan first.
```

Deploy help (you click, the agent explains)

```
I want to deploy this project on the free Render plan. Read render.yaml and README. Give me numbered steps with exactly what to click and which values to paste. Say what resets on every restart (the demo database) and why that is fine. Then tell me how to run /test-all against the live URL by setting BASE.
```

Review prompt (after every ticket) and quiz prompt

```
Review the last commit like a strict teacher. List: things I cannot explain easily, rule risks (real data, innerHTML, secrets, scraping), missing tests, edge cases, and anything that changes the sealed block format. Do not edit files.

Quiz me on <file or feature>. Ask 5 questions one at a time like a hackathon judge. Wait for my answers and correct me.
```

Phase B prompts (use after selection)

Start every Phase B prompt with this header, then add the feature text:

```
Read AGENTS.md, docs/SPEC.md, docs/UI_SPEC.md and docs/V2_FINAL_BLUEPRINT.md section 4. Additive only: new columns with defaults, new tables, new optional API fields. Never change the block canonical fields. New rules go in the rules registry with bn and en messages. All earlier tests must keep passing. New behaviour is behind the feature flag named below (env BHUMI_FEATURES). Write tests first and show them failing, then implement. Update docs/SPEC.md. Plan first and wait for approval.
```

Id	Prompt core (full text in docs/V2_FINAL_BLUEPRINT.md section 7)
B0	Refactor rules.evaluate() into a registry, add features.py and schema_version + migrate(), make the verifier ignore unknown fields. ZERO behaviour change; all tests pass unedited
F1 shares	holders table; optional area sold in the sale (put it in the deed text); rules SHARE_EXCEEDED (block) and CO_OWNERS_CONSENT (warn 50); partial sale updates shares; share bar UI
F2 heirs	SUCCESSION block kind; heirs table; HEIRS_SUM_INVALID when fractions do not sum to 1; succession form and family list
F3 lineage	survey_links walking CS, SA, RS, BS; checks area within 5 percent, share sums, holder links; SURVEY_SHARE_BREAK; four-column lineage UI
F4 map	polygon column; pure-Python overlap detection; BOUNDARY_OVERLAP; offline SVG mouza map colored by verdict; no external map service
F5 offline QR	QR carries the Ed25519 signature and office id; Web Crypto check against cached public keys; PWA shell; never show a cached verdict as current
F6 OCR	OpenCV clean-up + Tesseract Bangla + regex fields + confidence routing to an officer screen; only synthetic generated scans; honest accuracy numbers
F7 identity	demo identity token stored only as a hash; POA_SALE forces officer review; clearly labelled demo
F8 tax and flags	tax_paid_until, vested, waqf; rules TAX_DUE and VESTED_OR_WAQF; TAX as the 9th mutation check (update every "8 checks" text)
F9 SMS	/sms simulator, replies of at most 160 characters in Bangla
F10 data	tools/gen_synthetic_data.py: seeded, obviously fake names, loaded through Service; test that no value looks like an NID or phone number
F11 witness	three offices co-sign each anchor; verifier checks all signatures
F12 Postgres	optional DATABASE_URL backend behind the same Service, same triggers; no API change

8. Round 1 submission kit (four items, due Fri 16 Oct)

Item	Rule	Where it is	You still must
Concept note PDF	Maximum 4 pages: problem statement chosen, problem in Bangladesh, solution, architecture and tech stack, key features, countrywide plan, team and mentor details	docs/Bhumi_Lipi_Concept_Note.pdf (3 pages)	Fill team and mentor names; update "51 tests" to 53 if you keep that line; add Simple mode as a feature
Repository link	GitHub or GitLab, public or shared; README with run steps, data sources, AI tool disclosure; real commit history	the project zip	Push with commit history; list every AI tool and what it did
Demo video	Maximum 3 minutes	docs/DEMO_SCRIPT.md	Record it; add Simple mode (three-word verdict, big text) in the Citizen scene
Live link	Optional but recommended	render.yaml	Deploy; test on mobile data

3-minute video order

Time	Scene
0:00	Problem in two sentences (forged deeds, double selling). Home page
0:20	Role Sub-registrar: normal sale, seal and QR appear, plot locks
0:50	Fake seller blocked with a reason; second sale blocked
1:20	Citizen on a phone width: Simple mode, three-word verdict, big text, Listen if a Bangla voice exists
1:50	Drop genuine_deed.txt (Genuine) and tampered_deed.txt (Forged)
2:15	AC Land approves, waiting-days chip, plot unlocks. Integrity page and proof check
2:40	Attack lab: four stopped, one caught. Close with the logo

9. What to say and what not to say

Say	Never say
Tamper-evident signed hash chain; detected by the public anchor and the independent verifier	Tamper-proof, unhackable, blockchain-level security guaranteed
Status lock until mutation is approved	Two-phase commit (2PC)
No per-transaction fee; open-source stack	"80 to 90 percent cheaper" or any saving you did not measure
Prototype on synthetic data; designed to extend	"Ready for national rollout" or "tested at scale"
Phase B items are designed, not built	Showing a Phase B idea as if it works
Demo roles stand in for national digital ID	"Secure login"

Ten questions both students must answer (full list in docs/QA.md)

- Why not a public blockchain? (fees, speed, privacy; hash chain + signatures + public anchor give the proof)
- Is it tamper-proof? (No, tamper-evident; show attack 5 in the lab)
- Who holds the keys? (one key per office; production uses hardware keys)
- What if the database admin edits a row? (triggers block it; chain check and anchor catch it)
- What does the Merkle root prove? (a record existed when the root was published and history before it was not rewritten)
- How is a QR different from a fake QR? (the page reads the registry; a fake seal says FORGED)
- What about a valid old deed? (marked OUTDATED because a newer record exists)
- How do elders use it? (Simple mode: three words and shapes, big text, listen, print)
- What did the AI write and what did you change? (be ready to show your own diff)
- Walk me through one sale. (registrar enters, rules run, block signed, plot locked, mutation checks, AC Land approves, plot unlocks)

10. Risks and final checklist

Risk	Impact	Mitigation
Late or missing registration, mentors or fee	Everything is void	Confirm on the platform today (deadline was 6 Oct)
Unknown Judging Rubric	Effort in the wrong place	Download it; run /rubric-check
Students cannot explain code	Score drops, possible penalty	Daily /explain-back; rehearse QA.md
Late feature breaks the demo	Failed submission	Freeze Wed midnight; git tag round1-freeze; feature flags
Free host sleeps	Live link fails in the video or judging	Open it 5 minutes before; keep localhost ready; keep the video
Listen button not available on a device	Feature looks missing	Test on the demo phone; say it needs a Bangla voice
Wrong legal rule in a demo (shares, heirs, tax)	Credibility	Label as demo rules; ask a land-office clerk or law teacher
Real data slips in	Disqualification	/pre-submit scan; review screenshots and video

Checklist before Friday

- [ ] Team registered, fee verified, faculty and industry mentor assigned
- [ ] git tag round1-freeze exists; python -m unittest discover -s tests -t . shows 53 tests OK
- [ ] Browser test passes (tests/e2e_playwright.py)
- [ ] Live link works on a phone with mobile data
- [ ] Concept note has real names and no placeholders; at most 4 pages
- [ ] README: live link, run steps, data sources, every AI tool and what it did, team names
- [ ] Video under 3 minutes, plays logged out
- [ ] No real names, NID, phone numbers or land records anywhere (code, screenshots, video)
- [ ] Both students can explain every module and rehearsed the questions
- [ ] Submitted before noon on Friday with a confirmation screenshot

Honest self-assessment

Technically the project is strong and tested (53 automatic tests, a real-browser walkthrough, an independent verifier). Its biggest risks are not code: they are time (two days), whether both students can explain every file, and the unseen rubric. A score of 10/10 cannot be promised. The best way to protect your result is to keep the working version, finish the submission kit, and explain the project honestly, including what it does not do.